Experiment 14: write python program to Alpha-Beta Pruning Algorithm for Gaming Using Python

Aim:

To write a Python program to implement the Alpha-Beta Pruning algorithm for gaming.

Theory:

Alpha-Beta Pruning is an optimization technique used with the Minimax algorithm in Artificial Intelligence. It reduces the number of game-tree nodes that need to be evaluated without changing the final result.

- Alpha (α): The best value MAX can guarantee so far.
- Beta (β): The best value MIN can guarantee so far.
- Pruning: Stops exploring a branch when \(\alpha \geq \beta\).

Algorithm:

1. Start the program.
2. Initialize alpha to negative infinity and beta to positive infinity.
3. Evaluate the game tree recursively.
4. At MAX nodes, select the maximum value and update alpha.
5. At MIN nodes, select the minimum value and update beta.
6. If alpha is greater than or equal to beta, prune the remaining branches.
7. Display the best value.
8. Stop the program.

Python Program:

```
# Alpha-Beta Pruning Algorithm

def alphabeta(depth, node, alpha, beta, is_max, scores):
    if depth == 2:
        return scores[node]
    if is_max:
        best = float('-inf')
        for i in range(2):
            value = alphabeta(
                depth + 1, node * 2 + i,
```

```
                alpha, beta, False, scores
            )
            best = max(best, value)
            alpha = max(alpha, best)
            if alpha >= beta:
                break
        return best
    else:
        best = float('inf')
        for i in range(2):
            value = alphabeta(
                depth + 1, node * 2 + i,
                alpha, beta, True, scores
            )
            best = min(best, value)
            beta = min(beta, best)
            if alpha >= beta:
                break
        return best
scores = [3, 5, 2, 9]
result = alphabeta(
    0, 0, float('-inf'), float('inf'), True, scores
)
```

print("Best value:", result)

exp 14 alpha & beta.py - C:/Users/yadal/OneDrive/Documents/AI/exp 14 alpha & beta.py (3...

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```
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                depth + 1, node * 2 + i,
                alpha, beta, True, scores
            )
            best = min(best, value)
            beta = min(beta, best)

            if alpha >= beta:
                break

        return best
```

Output:

```
===== RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 14 alpha & beta.py
Best value: 3
>>>
```

Result:

Thus, the Alpha-Beta Pruning algorithm was successfully implemented in Python to find the optimal value in a game tree while reducing unnecessary node evaluations.